

Title.....	1
Author(s).....	1
ASSIGNMENT COVERSHEET	2
Database Design Assignment Part A.....	3
Assignment Teamwork Contract.....	9

Title

Database Design Assignment Part A– Spring 2024 - Apple App Store Case Study

Author(s)

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2. **Student ID and Name: 14510630 Abdulaziz Almehmadi**
3. **Student ID and Name: 25556693 Christopher Luu**

ASSIGNMENT COVERSHEET

UTS: ENGINEERING & INFORMATION TECHNOLOGY		
SUBJECT NUMBER & NAME 31271 Database Fundamentals	NAME OF STUDENT(s) (PRINT CLEARLY) Vyvyan Ran Almehmadi Abdulaziz Luu Christopher SURNAME FIRST NAME	STUDENT ID(s) 25465550 14510630 25556693
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NAME OF TUTOR Wesley	TUTORIAL GROUP Thursday 7:00PM	DUE DATE 01 September 2024
ASSESSMENT ITEM NUMBER & TITLE Database Design Assignment Part A– Spring 2024 Apple App Store Case Study		
I acknowledge that if AI or another nonrecoverable source was used to generate materials for background research and self-study in producing this assignment, I have checked and verified the accuracy and integrity of the information used. I confirm that I have read, understood and followed the guidelines for assignment submission and presentation on page 2 of this cover sheet. I confirm that I have read, understood and followed the advice in the Subject Outline about assessment requirements. I understand that if this assignment is submitted after the due date it may incur a penalty for lateness unless I have previously had an extension of time approved and have attached the written confirmation of this extension. Declaration of originality: The work contained in this assignment, other than that specifically attributed to another source, is that of the author(s) and has not been previously submitted for assessment. I have rewritten any material provided by AI or other nonrecoverable sources and where appropriate acknowledged their contribution. I understand that, should this declaration be found to be false, disciplinary action could be taken and penalties imposed in accordance with University policy and rules. In the statement below, I have indicated the extent to which I have collaborated with others, whom I have named. No content generated by AI technologies or other sources has been presented as my own work. Statement of collaboration: I would like to express my willingness to contribute to the Database Design Assignment Part A– Spring 2024 as a collaborator and equally take responsibility for completing my own part of the assignment. Signature of student(s) Ran Vyvyan, Abdulaziz Almehmadi, Christopher Luu Date 01 September 2024		

Database Design Assignment Part A

1. Case Study Overview

Apple App Store is an app marketplace that is exclusively compatible with the iPhone Operating System. The Apple website indicates that each user is required to create an Apple ID in order to enable the service. The necessary information is the user's full name, date of birth, email address or phone number, and password.

[Figure 1 - Create Apple ID Process]

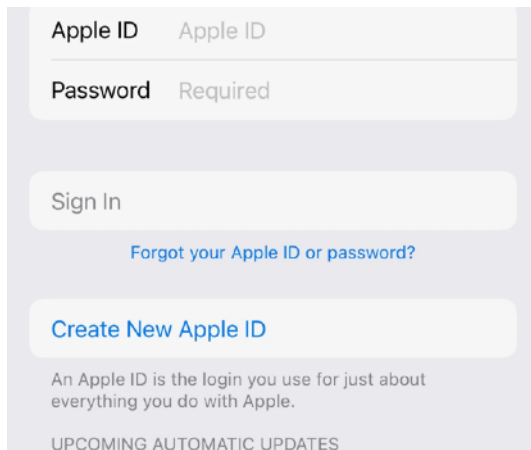


Figure 1 shows the initial 'Create Apple ID' screen. It features a header with 'Apple ID' and 'Apple ID' labels. Below this is a 'Password' field with a 'Required' status. A 'Sign In' button is present, along with a link for 'Forgot your Apple ID or password?'. A prominent 'Create New Apple ID' button is at the bottom, followed by explanatory text about Apple ID and a note about 'UPCOMING AUTOMATIC UPDATES'.

[Figure 2 - Create Apple ID Process]

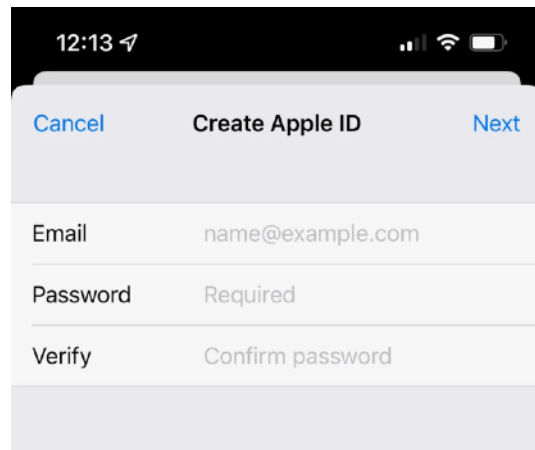


Figure 2 shows the 'Create Apple ID' form with a status bar at the top displaying '12:13'. The form has a header with 'Cancel', 'Create Apple ID', and 'Next' buttons. The main content area includes three input fields: 'Email' (with placeholder 'name@example.com'), 'Password' (with 'Required' status), and 'Verify' (with 'Confirm password' status).

[Figure 3 - Create Apple ID Process]

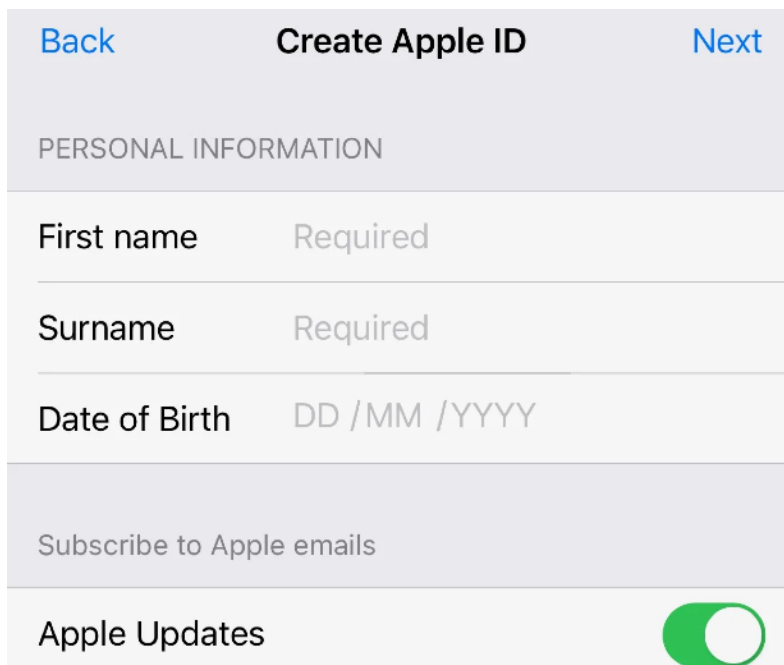


Figure 3 shows the 'Create Apple ID' form with a header containing 'Back', 'Create Apple ID', and 'Next' buttons. The form is titled 'PERSONAL INFORMATION'. It includes three input fields: 'First name' (with 'Required' status), 'Surname' (with 'Required' status), and 'Date of Birth' (with placeholder 'DD /MM /YYYY'). Below these is a 'Subscribe to Apple emails' section and an 'Apple Updates' section with a green toggle switch.

The payment method and billing information can be added after the previous step. However, it is optional, and if the user decides not to share the data, the user will not be charged until purchasing.

[Figure 4 & 5 - Add Payment options during Apple ID creation process]

12:06 84%

< Back Add Payment Done

PAYMENT METHOD
You won't be charged until you make a purchase.

Credit/Debit Card
Visa, MasterCard, American Express ✓

PayPal

Number Required

Expires MM / YYYY

CVV Security Code

12:06 83%

< Back Add Payment Done

PAYMENT METHOD
You won't be charged until you make a purchase.

Credit/Debit Card
Visa, MasterCard, American Express

PayPal ✓

Log In to PayPal

The street address, city, postal code, state and country are necessary for the billing address.

[Figure 6 - Adding the billing address]

Back	Create Apple ID	Submit
BILLING ADDRESS		
Street	15 Anon St	
Street	Zipton	
Street	Optional	
Postcode	Required	
Town	Nowheresville	

By providing a phone number, the system can verify the user's Identity. Finally, verify the email to activate the Apple ID.

[Figure 7 - Adding a phone number]



Phone Number

Please enter a phone number that can be used to verify your identity whenever you sign in on a new device or forget your password.

Country/Region +44 (GB) >

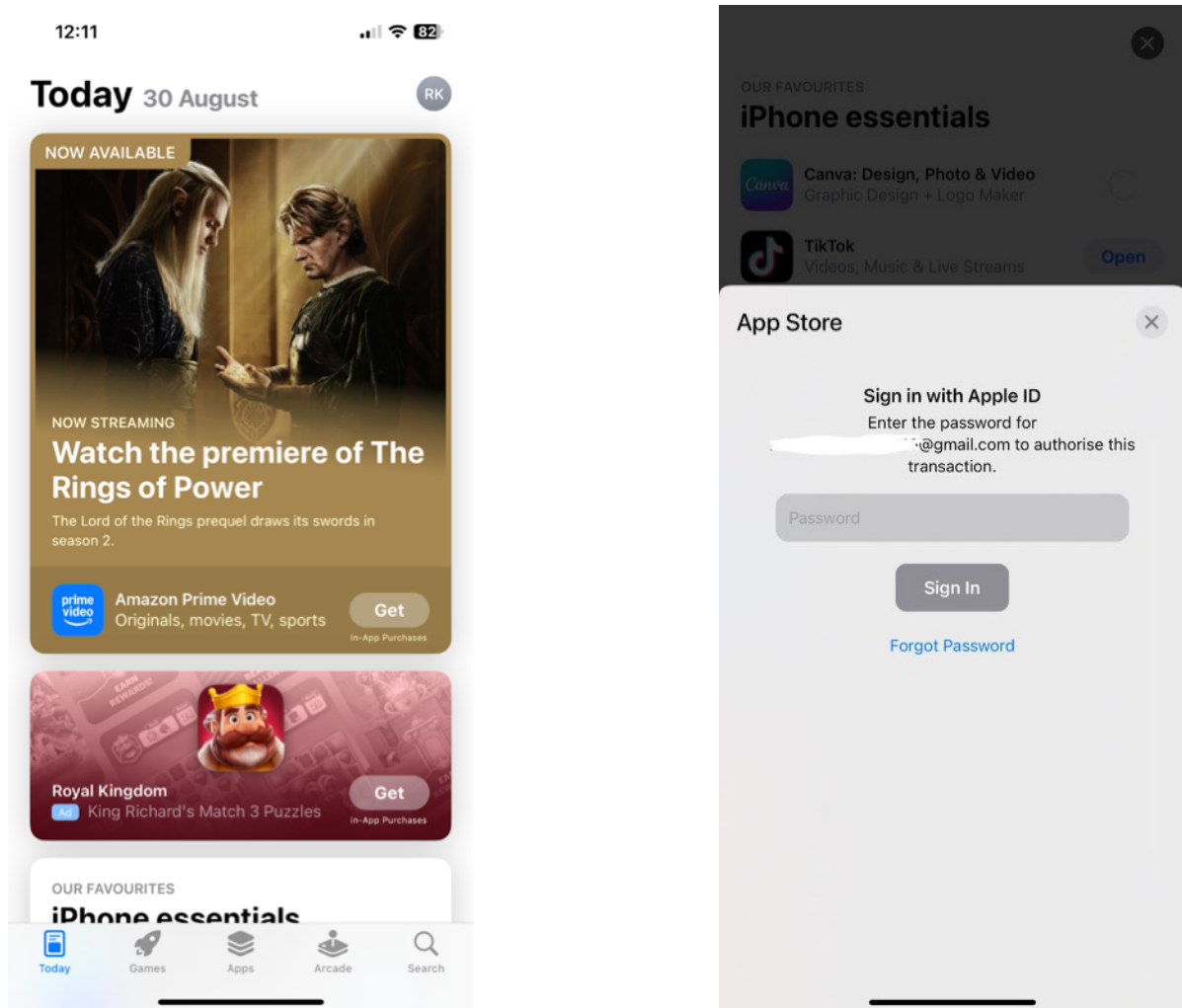
Phone 123 | 123456789

Searching the app in the store is a relatively straightforward and simple process. Once you have access to the store, use the search function to type in the application name or keywords. The system shows a dropdown list that matches the user inputs. The user can also explore the “Today”, “Games”, “Apps”, “Arcade”, and “Search” sections to find the app. The free apps are labelled with a “Get” button, and paid apps display their price as a button. [Figure 8 & 9]

To download the free application, click the button, and you are either asked for a Face ID or the password for your account to authorise the transaction. It has an option in case you forget the password as well.

To complete the purchasing or subscription process, the user is required to input the payment methods if the user has not added their payment methods. The system requires the user to choose the main payment system - Credit/Debit Card(Visa, MasterCard, American Express) or PayPal. Then, the user needs to provide the card number, expiration, CVV, and billing address if the user chooses the credit/debit card. If PayPal is the preferred method, the system asks the user to log into the PayPal account.

[Figure 8 & 9 - The Apple App Store and downloading process]



2. Database Functions

The database must be able to store user's information to generate an Apple ID to identify each user. The system must be able to verify users using phone numbers and email before active Apple ID. The system should allow the user to add his payment method for future purchases at any time.

In addition, it should store application details and provide the user with the ability to search using application names or keywords. The system should categorise applications into 'Games,' 'Apps,' and 'Arcade' and suggest today's trending applications.

Finally, the system should be able to verify the user before downloading applications using either a password or FaceID. The system should save the application's purchase details to create a bill. Even if the application is free, the system should save process information to be able to generate a page of downloaded applications.

3. Data Requirements

User: An individual who downloads, purchases, and uses apps from the app store.

User attributes: apple ID, device ID, email address, first and last name, phone number, date of birth, street, city, postal code, country, date of birth, and face ID.

Developers: An individual/organisation who produces applications for the app stores

Developer Attributes: developer ID, developer name, email address, phone number/business number, street, city, postal code, country, payment information (receive payments).

Payment: Payment information used for managing and storing payments, subscriptions, and purchase histories.

Payment attributes: apple ID, payment ID, payment method(credit, Apple Pay, PayPal), transaction amount, date and time, street, city, postal code, country, invoice number, refund_status, refund_date, and refund_reason.

Category: Various categories or genres of apps available in the App Store

Category Entity: Category ID, Category Name (Games, Productivity, Education), Category Description (apps belonging to the category).

Application: managed by developers and displayed to users.

Application attributes: application ID, Current version of the App, size, seller, compatibility, languages, age classification, price, contact information (phone number).

User Feedback: user feedback, suggestions and ratings for the application

User Feedback attributes: feedback ID, user ID, rating (1-5), review text, review title, developer response, app version (The version of the app being reviewed)

AppPricing: App fee encompasses pricing, subscription, and free application

AppPricing Attributes: app pricing ID, application ID, fee type (free, one-time purchase and subscription), price, subscription period, isTrialAvailable (boolean), trial period, auto-renewal (boolean)

4. Business Rules

- A user must have an iPhone and Apple ID to access the app store. If they have an iPhone and it's their first visit, they would have to sign up. Developers must also have a developer account.

- Users can only generate one Apple ID and not create multiple accounts to inflate app ratings and downloads.

- Users can only provide 1 review per application.

- Apps must obtain user consent before collecting personal data, and if collected must be informed of how their data will be used.

- Apps that cost money must be processed through Apple's payment system. The price of the app must be clearly stated in the app store listing.

- Developers are encouraged to respond to user reviews and feedback.

- Apps must provide user's a way to contact developers, handling inquiries, support requests and communication from Apple.

- Apps must have an appropriate age rating, see which languages are available, and the apps' compatibility with different devices and iOS versions.

- Apps must be able to see the developer's name, and application size, as well as the app's version history.

- Payments must be processed according to the user's billing address, including street, city, postal code, and country. This ensures correct tax calculation and compliance with local regulations.
- A user can download/purchase many apps from the store
- A user can redownload an app without providing a FaceID or password for the account.

Assignment Teamwork Contract

Part A

Assignment Title: Database Design Assignment Part A- Spring 2024 - Apple App Store Case Study

Student 1 Full Name & ID: Ran VYVYAN 25465550

Student 2 Full Name& ID: Abdulaziz Almeahmadi 14510630

Student 3 Full Name& ID: Christopher Luu 25556693

Assignment Marking Criteria: Part A: Case Study (5 marks)

Marking Criteria	Marks
1. Overview of the case study with the inclusion of forms	1
2. Outline functions that the database supports	1
3. Description of data requirements	1
4. Description of at least 10 business rules	2

Tasks/Activities Allocation:

Tasks	Student 1	Student 2	Student 3	Deadline	Status
1. Overview of the case study with the inclusion of forms	Ran			31 August 2024	Completed
2. Outline functions that the database supports			Abdulaziz	31 August 2024	Completed
3. Description of data requirements		Christopher		31 August 2024	Completed
4. Description of at least 10 business rules		Christopher		31 August 2024	Completed

Resources:

List the resources (books, articles, tools, software, etc.) that are used to complete the assignment.

Example: "I will design an ERD for ... website. Here is the link:"

The Apple ID create process information from the Apple official website

Link: <https://support.apple.com/en-au/108647>

The photography of the apple interface (Labelled as Figur

Link: <https://www.trustedreviews.com/how-to/how-to-create-an-apple-id-4221964>

Communication Plan:

Outline how the team will communicate (e.g., regular meetings, email, group chat) and how often.

Example: "The team will meet twice a week on Tuesdays and Thursdays at 4 PM"

The team communication was mainly on a group chat with a frequency of a few times per couple days. It changed based on the progression of the assignment.
One short meeting on Thursday before tutorial (6:30PM) to discuss about the assignment.

Conflict Resolution (if there was any):

Describe how conflicts or disagreements within the team will be handled.

Example: "If a conflict arises, the team will discuss the issue together and reach a consensus. If necessary, the instructor will be consulted."

In case a conflict arises, the team members will have a moment to discuss the issue in a respectful manner and try to resolve it within the team. If the issue is not resolved despite the previous effort, we might seek additional support from the tutor/third person.

Contribution and Accountability:

Specify how each member's contribution will be monitored and how accountability will be ensured.

Example: "Each member will submit weekly progress reports, and any member not contributing as agreed will be addressed in a team meeting."

Each member will be assigned a task and expected to complete it within a timely manner, and if there is an issue regarding contribution, a reminder will be given. A meeting can be scheduled to resolve the issue.

Signatures:

Student 1 Signature: Ran Vyvyan Date: 31 August 2024

Student 2 Signature: Abdulaziz Almeahmadi Date: 01 September 2024

Student 3 Signature: Christopher Luu Date: 01 September 2024